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Applied AI Research

Independent research through TechSoft Systems in agent reliability, memory provenance, evidence integrity, and local inference. Project leadership, hands-on integration, and AI-assisted development.

NVIDIA AI-Q Research Assistant

Ogden / Memory Workbench

PIPER Intent Gateway

Inference engineering

Agentic research and evidence integrity

NVIDIA AI-Q Research Assistant

Built a self-hosted AI-Q/NeMo research platform; analyzed 4.68M+ reported model tokens in selected retained studies and 1,448 calls across 45 diagnostic runs. Corrected a citation-accounting defect with 196 focused passing tests.

- Integrated NVIDIA AI-Q, NeMo Agent Toolkit, FastAPI, Dask, PostgreSQL, Next.js, vLLM, and Ollama for persistent research jobs, cited reports, and streaming progress.
- Built bounded cross-model evaluation with frozen evidence, positive/negative controls, deterministic source-passage identifiers, immutable inputs, and hash-chained execution journals.
- Designed typed request, budget, and evidence contracts to make research jobs traceable and replayable. Separated successful software execution from accepted research quality.
- Investigated source-binding, structured-output, and agent-handoff failures; tested source-first comparisons in independent contexts and identified limits in domain-specific controllers and generalization.
- Diagnosed a citation-accounting defect and restored 30 cited sources from a 77-source report in an exact-image replay; validated the repair with 196 focused passing tests and regression comparisons.

Active independent AI-assisted research, not commercial deployment. Token/call counts cover selected retained work, not lifetime totals. Software tests and execution journals do not establish factual correctness; bounded comparisons do not establish overall model superiority or reliable general research.

Intent classification and memory routing

PIPER Intent Gateway

Architected a native Hermes advisory plugin for intent and memory selection, deterministic contracts, monitoring, and human review; expanded a historical acceptance suite to 114 passing tests and audited 106 selected decisions.

- Led AI-assisted intent-routing research across early hybrid intake, Open-WebUI filters, and a later native Hermes advisory plugin. Historical gateway work included structured LLM output, deterministic policy mappings, bounded context, session isolation, timeout/fallback controls, and circuit breaking.
- Separated pre-work intake from post-work agent handoff. Historical routing contracts used schema validation, bounded repair, and fallback; model-assisted classification supplied hints rather than owning agent execution.
- Defined retrieval boundaries across conversation history, user profile, assistant operational memory, and authoritative durable knowledge. The executing agent retains planning and tool-use responsibility.
- Built gateway monitoring and review tooling with FastAPI/React, authenticated source-turn review, append-only corrections, revision-checked configuration, atomic replacement, backups, and rollback.
- Expanded the early gateway acceptance suite to 114 reported passing tests in May 2026. Re-reviewed 106 previously recovery-labeled decisions in September to separate memory-routing errors, classifier misses, internal turns, correct routes, and downstream recovery.

Independent R&D, documented May 2026 - Present. The native plugin is advisory, not autonomous memory enforcement. Historical packages do not establish feature parity across generations. The selected audit is failure analysis, not a general accuracy score; historical tests are not a full current-plugin passing run.

Proposal-only memory evaluation

Ogden / Memory Workbench

Evaluated 280 conclusions across 72 cases, completed after retries and serving recovery; produced 19 proposals for human review without mutating live memory.

- Designed and evaluated a proposal-only semantic memory reviewer with complete-artifact validation, idempotent imports, immutable model output, separate human edits, and append-only review events.
- Revalidated source records at review time and blocked approval on source drift. Kept live-memory application disabled, with auditable exports and human review separate from model-generated proposals.
- Established provenance-aware export boundaries: retained model conclusions are unreviewed, not human-approved training examples. Schema-valid output remains a proposal.
- Implemented exact deduplication that reduced a 200-conclusion sample to 111 review rows while retaining source identities and audit metadata.
- Researched governance distinctions between confidence, authority, and purpose, including privacy risks from cumulative profile inference; broader runtime enforcement remains planned.

Independent AI-assisted research. The 19 proposals are not 19 proven errors or applied fixes. Semantic clustering, a full memory compiler, and compaction concepts remain research/planned work, not completed features.

Local infrastructure and LLMOps

Inference & Platform Engineering

Convert base models to NVFP4 using NVIDIA Model Optimizer, validate and optimize the converted checkpoints, and deploy through vLLM. Work includes tensor parallelism, FP8 KV caching, MTP speculative decoding, prefix caching, chunked prefill, CUDA graphs, and model residency.

Integrate embedding/reranking services, local speech, and multimodal workloads. Preserve service state during container updates with verified backups, functional checks, and narrow rollback procedures.

- Designed reproducible inference comparisons using matched decoding settings, immutable model/runtime identities, bounded concurrency, and explicit timing definitions; retained failed attempts and workload-specific acceptance limits.
- Investigated tensor-parallel transport/IOMMU and GPU workspace failures, qualifying a single-GPU (TP1) serving configuration for Nemotron Super after unsuccessful dual-GPU (TP2) attempts. Distinguished per-sequence context limits from shared KV capacity and scheduling headroom.
- Evaluated role-aware embedding formats against public retrieval datasets and a separate hard-negative technical suite, informing workload-specific model selection rather than an overall quality ranking.

Selected Local Measurements

Nemotron Super 120B NVFP4

212.20 output tokens/sec

Warmed single-request technical workload; end-to-end rate across the configured generation, not decode-only throughput.

Nemotron Lightning 30B-A3B NVFP4

568.15 output tokens/sec

Mean of five warmed synthetic runs; 59.6% above the measured target-only control on the recorded runtime.

Qwen3.8-27B NVFP4

133.64-164.12 output tokens/sec

Median single-request decode rates on selected technical/coding prompt workloads; 914.20-1,130.73 aggregate output tokens/sec across eight concurrent requests.

Historical local-lab results use different workloads, timing definitions, and recorded runtime configurations. They are not direct cross-model rankings, universal model-performance claims, or guarantees on the current runtime. Detailed methodology remains in the private evidence record.